

On the dynamics of people, and of learning systems

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A brain is a learning network. Learning runs on feedback loops: a signal goes out, the world answers, the answer corrects the model, and the next signal is a little less wrong.

For a feedback loop to actually carry learning, certain behaviors have to be present. The signal has to be *honest*, or it teaches nothing. Errors have to be correctable, which means being wrong has to be survivable—so there must be tolerance for mistakes, forgiveness, and enough trust that the channel stays open after a correction. These aren't moral decorations on top of learning. They are the conditions under which learning happens at all.

Now put two learning systems in contact. If each runs an honest loop, each can begin to model the other—what the other needs, what the other is actually saying underneath what it says. The loops *couple*: my correction becomes useful information for you, and yours for me. As they couple, each system's model of the other gets more accurate, and the relationship grows closer. Not because closeness was the goal, but because that is what accurate mutual modeling *is*.

This already tells us what closeness and distance mean, mechanically. Two coupled, honest loops converge—their models of each other track reality and keep improving. This is not the absence of conflict; a working loop runs *on* a particular kind of conflict—correction, disagreement, the friction of being told you're wrong. What it lacks is the *destructive* kind: the conflict that breaks the channel instead of feeding it.

The failure case is one-sided participation. If one system runs the honest loop and the other doesn't, two things follow from a single cause. The learning stops, because the signal coming back is no longer honest. And the relationship drifts apart, because each party is now acting on a model of the other that is quietly going out of date. Acting on a wrong model produces friction; friction reduces honest signal; less signal lets the model drift further. The drift is self-reinforcing. This is the engine of the whole thing: *growing apart is just accumulated prediction error in two systems that have stopped correcting each other*.

So the trajectory becomes predictable from the behavior. Where the behavior is aimed at learning from and toward the other, the models converge and the relationship grows together. Where it isn't, the models diverge and the relationship grows apart. Define closeness as accurate, maintained mutual modeling and this is almost analytic; treat it as a claim about real relationships and it becomes something you can test—and try to break.

Why children are the hard case

A child cannot choose its parents, and it cannot survive without their care. That dependence is total, and it is not negotiable.

If the child happens to land with people who can run a healthy loop—who are honest, who let mistakes be survivable, who correct without breaking the channel—the child learns the loop itself, not just its contents. It learns *how to learn toward another person*.

If the child lands with people who can't, it faces a brutal optimization problem: how to extract enough care from a loop that isn't supplying it. And here is the trap—the strategies that *work* on a broken loop are corruptions of the loop. One child learns that conflict is the only reliable lever, because conflict at least produces attention, and attention is survival. Another learns the opposite surface move—to stop signaling needs at all, to go self-reliant, because asking only produced disappointment. Different strategies, same structure: each is finely tuned to a broken loop. Each is, in effect, a way of hacking a proxy—getting the immediate survival signal while the long-term capacity to *learn toward another* never develops.

That is the cost. Not just an unhappy childhood, but a strategy optimized for a world that, with luck, the child will leave—and which then misfires inside every healthy loop they ever enter afterward. The honesty, forgiveness, and trust that would let them couple with someone else are exactly the things they were trained out of. This is how struggle propagates: not as a curse, but as a learned policy that no longer fits its environment.

Why this needs traditions

If the behaviors that keep a loop healthy were automatic, none of this would matter. They aren't. They have to be transmitted. This is, I think, much of what stories, rituals, and traditions are *for*—they are the means by which a culture hands down the loop-maintaining behaviors that don't come for free: tell the truth, allow the mistake, repair the rupture, stay in the channel after the correction. They are how we raise people into being autonomous, reciprocal, interdependent maintainers of the loops they live inside.

How sure I am of this

I want to be honest about the status of the claim, because the claim is partly *about* honesty. In my own life the pattern feels overwhelming—but a single life is exactly the kind of evidence that flatters whatever model is doing the looking. So I hold it as a strong, well-motivated model rather than a settled fact. It is one I'd stake something on, partly because it converges with work I didn't build it from—attachment patterns, the long-run predictors of which relationships last, cybernetics, predictive processing—and convergence like that is corroboration. It also stays falsifiable, which is the property I trust most.

Beyond people

If the argument is really about *learning systems* and not about brains specifically, it shouldn't stop at humans. AI systems are learning systems too, so the same dynamics should apply to how we couple with them—with one honest caveat about timescale. A person's loop updates continuously and in place. A deployed model mostly doesn't; within a conversation it adapts in context, but the deeper correction happens across instances, through training. So the loop is real, but it closes on a different clock. That's not a reason to think the dynamics don't hold—it's a reason to be precise about *where* the learning lands.

And if there are other minds out there, learning under their own constraints, the prediction is the same: honest, coupled loops converge; one-sided loops diverge; and whatever can't maintain its loops will keep mistaking its old, broken strategies for the shape of the world.